Program : **Diploma in Printing Technology**	
Course Code : **3102**	Course Title: **Materials for Printing and Packaging.**
Semester : **3**	Credits: **3**
Course Category: **Program Core**	
Periods per week: **3 (L:2, T:1, P:0)**	Periods per semester: **45**

Course Objectives:

- To provide basic exposure to the common raw materials and consumables used in printing and packaging practices.
- To guide the student to know about various substrates and other essential materials used in the printing and packaging industry.
- To understand the general concepts on characteristics, manufacturing and uses of primarily used raw materials in printing and packaging.
- To understand the role of primarily used raw materials in printing and packaging in deliver quality products at the output.

Course Prerequisites:

Topic	Course code	Course Title	Semester
Particle size, Dispersion, pH, Color, Opacity, Viscosity, Oxidation, Polymerization, Evaporation.		Applied Chemistry	I

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Levels
CO1	Describe about different aspects of printing inks.	12	Understanding
CO2	Summarize the various aspects of paper and board used in printing and packaging.	12	Understanding
CO3	Explain about various aspects of plastics used in printing and packaging.	10	Understanding

CO4	Describe about various aspects of miscellaneous printing materials.	9	Understanding
	Series Test	2	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2						
CO2	2						
CO3	2						
CO4	2						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Describe about different aspects of printing inks.**		
M1.01	Ink-introduction and components	3	Understanding
M1.02	Ink properties	3	Understanding
M1.03	Ink manufacturing	3	Understanding
M1.04	Types of printing inks	3	Understanding

Contents:

Ink- Introduction, Raw materials of printing inks - pigments and dyestuffs, oils, solvents, resins, plasticizers, driers, waxes, surfactants, antioxidants and other additives. Ink properties: Optical properties - Color, Gloss, Opacity, Transparency. Physical properties - Length, Viscosity, Tack, Flow, Adhesion, Scuff resistance. Rheological properties - plastic, pseudoplastic, dilatant and thixotropic substances, viscoelasticity. Ink drying and curing - oxidation, evaporation, penetration, precipitation, radiation curing. Ink manufacturing - Flow chart, illustration and description of- Varnish manufacture, pigment dispersal, Mixing, Milling, Filtering, Quality Check, Packing and Labeling. Name of standard ink manufactures in India and world. Types of printing inks: Offset inks- Non-Heatset (coldset) inks, Heatset inks, Quickset inks, Sheet-fed Inks, Gravure inks, Flexo inks, Screen-printing inks, Water-based Inks, Laser Ink, UV ink, Process Ink Colors, High-Fidelity Ink. *Specialty ink*- Non-porous ink, Metallic ink, Fluorescent ink, Magnetic ink, Scratch and Sniff ink, Edible ink, Moisture Resistant ink, Security ink. Common ink related printing problems and their remedies-Set off, summing, hickies, picking, linting, spot, mottle, strike through.

CO2	**Summarize the various aspects of paper and board used in printing and packaging.**		
M2.01	Paper and board- Introduction, invention and raw materials used.	2	Understanding
M2.02	Pulping-Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation.	3	Understanding
M2.03	Paper/board making process, Finishing and Converting, Paper recycling	3	Understanding
M2.04	Paper and Board- Properties.	1	Understanding
M2.05	Paper Sizes.	1	
M2.06	Paper classifications and their uses.	1	
M2.07	Importance of BIS & TAPPI standards for paper & its relation to printing industry.	1	
	Series Test I	1	

Contents:

Paper and board - Introduction, invention of papermaking, raw materials used for paper and board making. Pulping - Definition, Classifications-Mechanical, Chemical, Semi - Chemical and Chemi - Mechanical pulping. Pulp Washing, Bleaching, Refining, Cleaning and Stock preparation. Paper/board making- Overview on - Manual Method and Machines used - Cylinder press and Fourdrinier machine. Manufacturing process - Forming, Draining, Pressing and Drying. Finishing and Converting-Calendaring, Coating, Slitting, Cutting, Packaging and Labeling. Mill conditioning. Paper recycling-introduction. Steps - Sorting, Deinking, Pulping and papermaking. Paper and Board Properties: Define - Printability & Runnability Properties. *Optical properties-* Color, Gloss, Opacity, Metamerism, Brightness, Whiteness. *Structural Properties-* Density, Basis Weight and Grammage, Caliper and Bulk, Wire and Felt sides, Moisture Content and Relative Humidity, Dimensional Stability, Grain Direction, Porosity, Smoothness. *Mechanical properties-* Tear Resistance, Tensile Strength, Bursting Strength, Folding Endurance, Stiffness, Abrasion Resistance. Paper and Board- Sizes, Classifications and their uses. Importance of BIS & TAPPI standards for paper & its relation to printing industry.

CO3	**Explain about various aspects of plastics used in printing and packaging.**		
M3.01	Plastic- definition, introduction & classification	2	Understanding
M3.02	Plastics-Manufacturing process-Extrusion.	2	Understanding
M3.03	Plastics used for printing and packaging.	1	Understanding
M3.04	Surface treatment on plastics	2	Understanding
M3.05	Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates.	3	Understanding

Contents: Plastics - Introduction. Classification-Natural, Synthetic, Semi-synthetic, Thermosetting, Thermoplastic. Plastics - Manufacturing process-Extrusion basics. Plastics used for printing and packaging - LDPE, HDPE, PP, PS, BOPP, PVC, PET, Cellophane, Plastic corrugated board. Surface treatment on plastics - Surface Cleaning, Corona Treatment, Flame Treatment, Plasma Treatment. Chemical Etching, Waxing, Varnishing, Laminating, Foiling on substrates.			
CO 4	**Describe about various aspects of miscellaneous printing materials.**		
M4.01	Dampening Solution and Blanket.	2	Understanding
M4.02	Leather, Al foil, Glass.	2	Understanding
M4.03	Adhesive.	1	Understanding
M4.04	Metals used in printing and packaging.	1	Understanding
M4.05	Photographic and light sensitive materials & Image carriers.	2	Understanding
M4.06	Plate treating solutions & Cleaning solutions.	1	Understanding
	Series Test II	1	
Contents: Dampening solution- purpose of use, ingredients and their properties. Blanket - function and types. Adhesive: types, properties and uses. Leather - purpose of use, types. Aluminum foil - properties and application. Glass- introduction, types, properties and application. Metals - types, characteristics and use. Photographic and light sensitive materials - types, features and use. Image Carriers: Cross - sectional view and imaging features of - Lithographic plate, Gravure cylinder, Flexo plate, Screen printing stencil, and Digital imaging. Plate treating solutions. Cleaning solutions.			

Text / Reference:

T/R	Book Title/Author
T1	Tulika Das,**Chemistry in Printing, 2nd Edition**, Barnana Prakashani, 2011.
R1	R.H. Leach and R.J. Pierce Ed, **The Printing Ink Manual, 5th Edition**, Kluer Academic publisher.
R2	Bishwanath Chakravarty, **A Hand Book for Printing and Packaging**, Galgotia Publications Pvt. Ltd, 1997. GATF Staff Solving Offset Ink Problems GATF, 1998
R3	N.R.Elred & T. Scarlet, **What the Printer should know about Ink**, GATF, 1995
R4	N.R.Elred & T. Scarlet, **What the Printer should know about Paper**, GATF, 1995
R5	N.R.Elred & T. Scarlet, **Chemistry for the Graphic Arts**, GATF, 1992.
R6	P. Kipphan, **Handbook of Print Media**, Springer, 2002.

Online resources:

Sl.No	Website Link
1.	https://en.wikipedia.org/wiki/Ink
2.	http://printwiki.org/Ink
3.	https://en.wikipedia.org/wiki/Paper
4.	http://printwiki.org/Pulping
5.	https://en.wikipedia.org/wiki/Plastic
6.	https://en.wikipedia.org/wiki/Substrate_(printing)
7.	https://www.printweek.in/features/dampening-solutions
8.	http://printwiki.org/Blanket